UNIFIED CYBER OLYMPIAD

S Y L L A B U S

Class - 2

Mental Ability : Numbers, Concept of 0, Order of numbers, Addition, Subtraction, Understanding of multiplication and division, Measurement, Time, Money.
Reasoning : Odd One Out, Embedded Figures, Series, Similar shapes, Analogies, Mathematical reasoning, Analytical reasoning.
Computers : Introduction to computers, General Information about Computers, Uses of computers, Parts of a computer, Mouse and Keyboard, MS-Paint.
English : Vocabulary, Grammar and Correct usage.

Class - 3

Mental Ability : Numbers upto 10000, Place value, Round off numbers, Expanded notation, Sum or difference of two numbers between 0 and 10000, Inverse relationship of multiplication and division, Solve simple problems involving multiplication and division, Special properties of 0 and 1 in multiplication and division, Identifying fractions, Simple unit conversions, Simple problems, Involving a functional relationship between two quantities, estimate and measure the length, liquid volume, and weight/mass of given objects, Solid shapes, Open and closed figures, Time, Money.
Reasoning : Analogy, Series completion, Simple coding, Alphabet test, Adding figures, Grouping of identical figures, Odd one out, Incomplete pattern, Inserting missing numbers, Analytical reasoning.
Computers : Introduction to Computers, Keyboard, Operating Systems, Parts of Computer, Uses of Computer, Input and Output Devices, Introduction to Internet, MS-Paint, Introduction to MS-Word.
English : Vocabulary, Grammar and Correct usage.

Class - 4

Mental Ability : Numbers (Indian & International), Place value, Round off numbers, Compare whole numbers and decimals to two decimal places, Different interpretations of fractions, Write tenths and hundredths in decimal and fraction notations, Applications on addition, Subtraction, multiplication and division, Concept of factors and multiples, Measurement, Time, Money, Data handling.
Reasoning : Analogy, Series completion, Simple coding, Alphabet test, Subtracting figures, Grouping of identical figures, Odd one out, Incomplete pattern, Inserting missing numbers, Analytical reasoning.
Computers : Introduction to computers, More about Windows, Characteristics and Evolution of Computers, Parts of computer, Input and Output Devices, Hardware, Software, Notepad, Introduction to Logo, Logo Commands, Writing and Arithmetic with Logo, Introduction to Microsoft Word, Introduction to Internet and its uses. Computer Networks, MS-Paint.
English : Vocabulary, Grammar and Correct usage.

Class - 5

Mental Ability : Estimate and manipulate very large (e.g., millions) and very small (e.g., thousandths) numbers, Finding HCF & LCM, Identify and represent on a number line, add, subtract, multiply, and divide with decimals, Solve simple problems, including ones arising in concrete situations, involving the addition and subtraction of fractions and mixed numbers, Measurement, Time, Money.
Reasoning : Analogy, Series completion, Simple coding, Alphabet test, Mirror images, Folded in, Grouping of identical figures, Odd one out, Embedded figures, Mathematical reasoning, Analytical reasoning, Jumbled words, Forming new words.
Computers : Introduction to Computers, Data Storage Devices, More about Windows, More Logo Commands, More Commands in Logo, Editing Text in MS Word, Formatting in MS Word, Introduction to MS Power Point, Working with text in MS Power Point, Multimedia, Input & Output Devices, Hardware, Software, Memory – Primary & Secondary Memory, MS-Paint, Internet, Computer Networks.
English : Vocabulary, Grammar and Correct usage.

Class - 6

Mental Ability : Perimeter and Area (Grandmother's Quilt), Time (Racing Seconds), Common Factors (Animal Jumps), Maps and Locations, Data Through Pictures, Patterns in Mathematics, Lines and Angles, Number Play, Prime Time, Fractions.
Reasoning : Analogy, series completion, simple coding, alphabet test, mirror images, formation of figures, opened out, grouping of identical figures, odd one out, mathematical reasoning, analytical reasoning, Jumbled words, Forming new words.
Computers : Introduction to Computers, Computer Generations, Computer Peripherals, MS Word: Features, Creating Table Working with objects in MS Word, Advanced features, MS Power point Presentations: Introduction, creating a presentation, Backgrounds and slide organization, Adding Graphics and organization charts, Enlivening a Presentation, Microsoft Excel, Internet & E-mail, Flowcharting, MS-Windows, Introduction to QBasic, Hardware, Software, Input & Output Devices, Memory and Storage Devices.
English : Vocabulary, Grammar and Correct usage.

Class - 7

Mental Ability : Fractions, Playing with Constructions, Symmetry, The Other Side of Zero, Large Numbers Around Us, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers, Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions.

Reasoning : Analogy, series completion, coding and decoding, alphabet test, mirror images, formation of figures, paper cutting, grouping of identical figures, odd one out, analytical reasoning, direction sense test, mathematical reasoning, Jumbled words, Forming new words.

Computers : Introduction to computers, Evolution of Computers, Microsoft Excel Editing cell contents in MS Excel, Formatting Data in MS Excel, Formulae in MS Excel, Advanced features of MS Excel, Creating Charts, Internet, Macromedia Flash, Drawing in Flash, Q Basics, Q Basic statements, Memory & Storage Devices, Using Windows, MS-Word, MS-PowerPoint, Networking.

English : Vocabulary, Grammar and Correct usage.

Class - 8

Mental Ability : Geometric Twins, Operations with Integers, Finding Common Ground, Another Peek Beyond the Point, Connecting the Dots, Constructions and Tilings, Finding the Unknown, A Square and A Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play, We Distribute Yet Things Multiply, Proportional Reasoning - 1.

Reasoning : Analogy, Series completion, Coding and decoding, Alphabet test, Mirror images, Paper folding, Grouping of identical figures, Odd one out, Analytical reasoning, Direction sense test, Mathematical reasoning, Cubes and dice, Locating hidden words, Jumbled letters, Formation of new words.

Computers : Review of computer basics, Computer Viruses - Types, Prevention, Antivirus, Updating virus definitions, Q Basic - Looping statements, Graphics and Sound, Internet - Searching Information, Communication on the Internet, Flash - Tools, Understanding timeline, Frames and Key frames, Layers and Animation, HTML Basics - Introduction to HTML, HTML Elements - Tags and Attributes, Structure of an HTML Document, Creating an HTML Document, Line Break
 Tag, Paragraph <P> Element, Horizontal Rule <HR> Element, Comment Element, HTML Text - Text Elements, Address Tag, Pre Tag, Center Tag, Heading Tag, Font Tag, Base Font Tag Structure, Commands, (Tables and Linking), MS Access - Database and Fields, Modifying table designs. Fundamentals of computers, MS-PowerPoint, MS-Excel, Memory & Storage Devices, Basics of Cyber crimes, Cyber Laws, Operating Systems.

English : Vocabulary, Grammar and Correct usage.

Class - 9

Mental Ability : Fractions in Disguise, The Baudhayana-Pythagoras Theorem, Proportional Reasoning - 2, Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area, Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials, The World of Numbers, Exploring Algebraic Identities, I'm Up & Down, and Round & Round (Circles).

Reasoning : Analogy, Series completion, Coding and decoding, Alphabet test, Mirror images, Blood relations, Grouping of identical figures, Odd one out, Analytical reasoning, Direction sense test, Mathematical reasoning, Cubes and dice, Locating hidden words, Jumbled letters, Formation of new words.

Computers : Review of Computer basics, Computer System - Evolution and History of computers, Generation, Electronic Computers, Hardware and Software, Language, Terms related to open and non-open source software, Operating system - Windows XP, MS Word, MS Front Page, Visual Basic - GUI, VB Window, Farm object, Common controls, Photoshop - Tools, Layers, Filters, Networking and E-commerce, Algorithms and Flowcharts in Programming, Animals, HTML, MS-Excel, MS-PowerPoint, Input & Output Devices, Memory & Storage Devices, Flash CS6, Communication Technology, Operating Systems.

English : Vocabulary, Grammar and Correct usage.

Class - 10

Mental Ability : Triangles, Quadrilaterals, Circles, Heron's Formulae, Surface Area and Volume, Statistics, Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations, Arithmetic Progression.

Reasoning : Analogy, Series completion, Coding and decoding, Alphabet test, Mirror images, Blood relations, Grouping of identical figures, Odd one out, Embedded figures, Problems based on figures, Analytical reasoning, Direction sense test, Mathematical reasoning, Cubes and dice, Locating hidden words, Jumbled letters, Formation of new words.

Computers : Review of Computer basics, MS Powerpoint, MS Excel, Visual basic - Tool box, Working with data, Selection and looping, C++ - Structure, Variable, Constants, Standard I/O. Operators, Control structure, Internet - Hisotry, Internet services, Web - browsers, MS Access - Creating new Database, Tables, Forms, Queries and reports, Societal Impact of Information Technology - Security and Integrity of information, HTML - Creating Lists and Inserting Images, Linking - Creating Hyperlinks using <A> Tag, Link Within a Single Page, Creating Images as Link, Fundamentals of computer, Operating System, Word Processing Tool, Networking, Internet, Database Management System (Functions, Types, MS-Access).

English : Vocabulary, Grammar and Correct usage.